

Geometry from Spline Interpolation

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Abstract

To come.

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Chapter 1

Introduction

To come.

Chapter 2

Methods

2.1 General Splines

Let a collection of $n+1$ points (x_i, y_i) for $i = 0, 1, \dots, n$ approximate the function $y = f(x)$ over the domain $[x_0, x_n]$. In total, there are $n+1$ points and n intervals. Let $S(x)$ be the piecewise continuous spine that threads through each of the points (x_i, y_i) . For $x \in [x_i, x_{i+1}]$ and $i = 0, 1, \dots, n-1$, the spline takes on n interval definitions as follows:

$$S_i(x) \triangleq a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 + \dots + z_i(x - x_i)^p. \quad (2.1)$$

where constant coefficients¹ multiply successive powers² of $(x - x_i)$, summed to an arbitrary polynomial power p . This is referred to as the **interval spline definition** to polynomial order p . Below we consider linear splines ($p = 1$), quadratic splines ($p = 2$), and cubic splines ($p = 3$).

2.2 Linear Splines

For linear splines, the interval spline definition is

$$S_i(x) \triangleq a_i + b_i(x - x_i) \quad \text{for } x \in [x_i, x_{i+1}] \text{ and } i = 0, 1, \dots, n-1. \quad (2.2)$$

For any collection of points (x_i, y_i) , all coefficients a_i and b_i for $i = 0, 1, \dots, n-1$ can be found using the **zeroth derivative continuity constraint**, which creates a spline that connects to each point at the beginning and end of each interval, thus

$$S_i(x_i) = y_i \quad \text{and} \quad S_i(x_{i+1}) = y_{i+1} \quad \text{for } i = 0, 1, \dots, n-1. \quad (2.3)$$

Here we see there are two constraints per interval i and two unknowns, a_i and b_i , per interval i . The unknowns can be found directly by evaluating the interval spline definition with each of the

¹Here the z_i is arbitrary, meant to denote the last coefficient, but not necessarily the twenty-sixth coefficient.

²The powers are non-negative integers.

two constraint equations. In turn, these are, for $i = 0, 1, \dots, n - 1$,

$$S_i(x_i) = a_i + b_i(x_i - x_i) \stackrel{\text{set}}{=} y_i \implies a_i = y_i, \text{ and} \quad (2.4)$$

$$S_i(x_{i+1}) = a_i + b_i(x_{i+1} - x_i) \stackrel{\text{set}}{=} y_{i+1} \implies b_i = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}. \quad (2.5)$$

No additional continuity constraints can be created since the linear spline has only C^0 continuity, meaning no derivatives of the function $S_i(x)$ are continuous.

Example 1.

Let $y = f(x) = \sin(\pi x)$ for $x \in [0, 1]$ be approximated by a linear spline and a collection of three points equally spaced on the domain. Note $n + 1 = 3$, thus $n = 2$, indicating three points and two intervals. Table 2.1 summarizes the spline data. Figure 2.1(a) shows the sin function and the linear spline interpolations.

Table 2.1: Linear spline approximation of $y = f(x) = \sin(\pi x)$ using three points and two intervals.

i	x_i	y_i	$\implies$	a_i	b_i	$\implies$	$S_i(x)$
0	0/2	0		0	2		$2x$
1	1/2	1		1	-2		$1 - 2(x - 1/2)$
2	2/2	0		—	—		—

Example 2.

Repeat the previous example, but now with a collection of five points equally spaced on the domain. Thus $n + 1 = 5$, thus $n = 4$, indicating five points and four intervals. Table 2.2 summarizes the spline data. Figure 2.1(b) shows the sin function and the linear spline interpolations.

Table 2.2: Linear spline approximation of $y = f(x) = \sin(\pi x)$ using five points and four intervals.

i	x_i	y_i	$\implies$	a_i	b_i	$\implies$	$S_i(x)$
0	0/4	0		0	$2\sqrt{2}$		$2\sqrt{2}x$
1	1/4	$\sqrt{2}/2$		$\sqrt{2}/2$	$4 - 2\sqrt{2}$		$\sqrt{2}/2 + (4 - 2\sqrt{2})(x - 1/4)$
2	2/4	1		1	$2\sqrt{2} - 4$		$1 + (2\sqrt{2} - 4)(x - 2/4)$
3	3/4	$\sqrt{2}/2$		$\sqrt{2}/2$	$-2\sqrt{2}$		$\sqrt{2}/2 - 2\sqrt{2}(x - 3/4)$
4	4/4	0		—	—		—

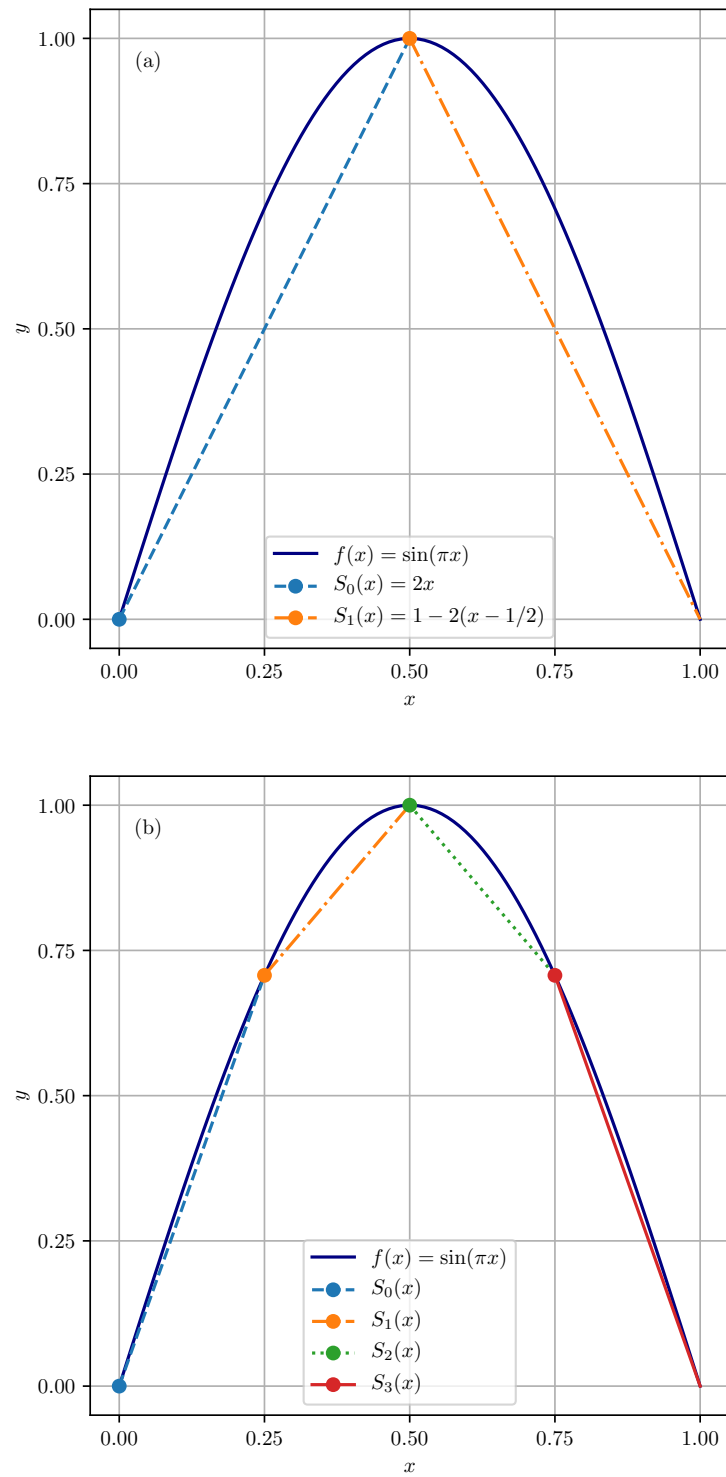


Figure 2.1: Linear spline example (a) with three points and (b) with five points, see Table 2.2 for individual definitions of $S_i(x)$. Reference: `fig/linear_spline.py`

2.3 Quadratic Splines

For quadratic splines, the interval spline definition is

$$S_i(x) \triangleq a_i + b_i(x - x_i) + c_i(x - x_i)^2 \quad \text{for } x \in [x_i, x_{i+1}] \text{ and } i = 0, 1, \dots, n-1. \quad (2.6)$$

2.4 Cubic Splines

Cubic splines are useful because of their inherent smoothness. The cubic spline, its first derivative, and its second derivative are all continuous. Starting with the cubic spline's third derivative, discontinuity arises.

Let a collection of $n + 1$ points (x_i, y_i) for $i = 0, 1, \dots, n$ approximate the function $y = f(x)$ over the domain $[x_0, x_n]$. In total, there are $n + 1$ points and n intervals. Let $S(x)$ be the piecewise continuous spline, cubic in this case, that threads through each of the points (x_i, y_i) . The spline takes on n interval definitions as follows:

$$S_i(x) \triangleq a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 \quad \text{for } x \in [x_i, x_{i+1}] \text{ and } i = 0, 1, \dots, n-1. \quad (2.7)$$

The n foregoing definitions, with the coefficients a_i, b_i, c_i, d_i , give rise to $4n$ (currently unknown) parameters used to define the complete spline $S(x)$. Note that maintain its cubic nature, the spline must have $d_i \neq 0$.

Next we endow the spline with constraints on continuity and differentiability, which allow the $4n$ unknown parameters to be solved. First, function continuity requires that the spline, on a particular interval, thread through collection of points (x_i, y_i) and the beginning and end of each interval. Thus,

$$S_i(x_i) = y_i \text{ and } S_i(x_{i+1}) = y_{i+1} \quad \text{for } i = 0, 1, \dots, n-1. \quad (2.8)$$

This is the **continuity constraint**, which gives rise to $2n$ conditions³ and creates a spline that is piecewise continuous. Next, consider two **differentiability constraints**, which state that the first and second derivatives at a point x_i are continuous. That is, the first and second derivatives approaching a point x_i from the interval to the left side of the point must match the respective first and second derivatives approaching the point x_i from the interval to the right side of the point:

$$S'_{i-1}(x_i) = S'_i(x_i) \text{ and } S''_{i-1}(x_i) = S''_i(x_i) \quad \text{for } i = 1, 2, \dots, n-1. \quad (2.9)$$

Because they apply only to the interior points (and not the endpoints x_0 and x_n), the differentiability constraints give rise to $2(n-1)$ constraints.⁴

³Two conditions per interval times n intervals yields $2n$ conditions.

⁴Two conditions per interior point times $n-1$ interior points yields $2(n-1)$ conditions.

Finally, two more conditions are necessary to completely define the spline $S(x)$. The two conditions are obtained by specifying differentiability conditions on the two spline endpoints x_0 and x_n . Frequently, the first derivative is set to the known value of the function's derivatives at the endpoints,

$$S'_0(x_0) = f'(x_0) \text{ and } S'_{n-1}(x_n) = f'(x_n). \quad (2.10)$$

This is defined as a **clamped spline**. Alternatively, a **natural spline** defines the second derivatives to be zero at the endpoints,

$$S''_0(x_0) = 0 \text{ and } S''_{n-1}(x_n) = 0. \quad (2.11)$$

Finally, a **not-a-knot** spline requires no additional constraints on the endpoint derivatives,⁵ but requires that the *third* derivative be continuous at x_1 and x_{n-1} .

⁵This has no effect on the original continuity constraint on the endpoints, that $S_0(x_0) = y_0$ and $S_{n-1}(x_n) = y_n$.